Bandwidth utilization is the wise use of available bandwidth to achieve specific goals.

Efficiency can be achieved by multiplexing;

MULTIPLEXING

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic.

- **Frequency-Division Multiplexing**
- **Wavelength-Division Multiplexing**
- **Synchronous Time-Division Multiplexing**
- **Statistical Time-Division Multiplexing**

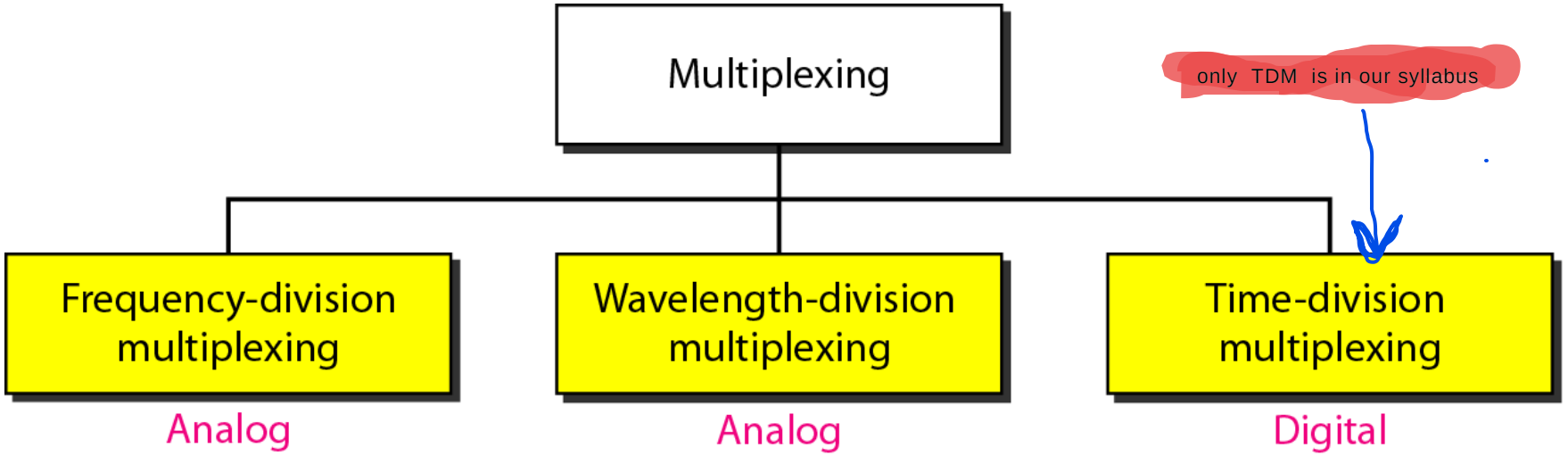
MULTIPLEXING

Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

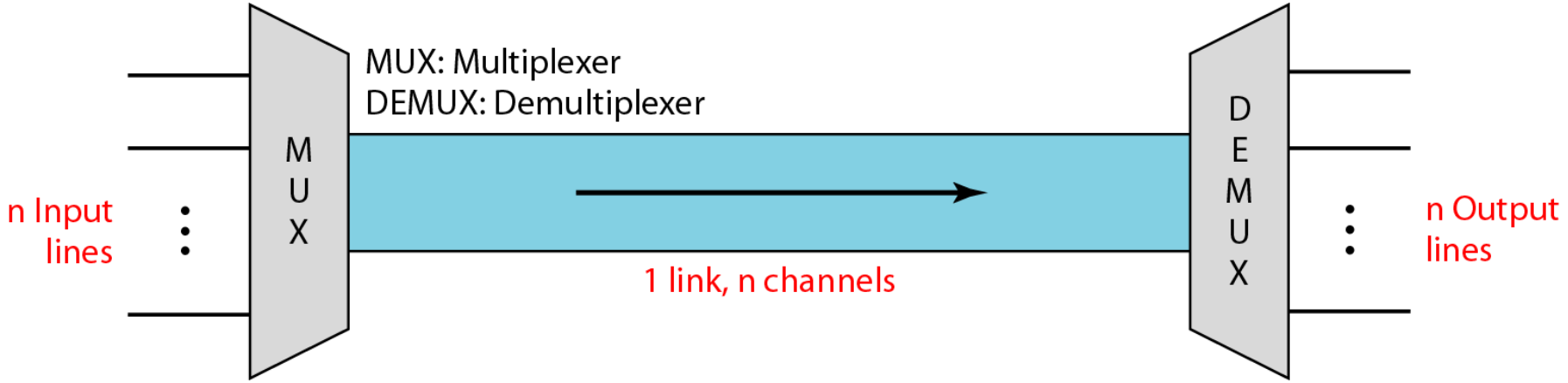
A **Multiplexer** (MUX) is a device that combines several signals into a single signal.

A **Demultiplexer** (DEMUX) is a device that performs the inverse operation.

Categories of multiplexing

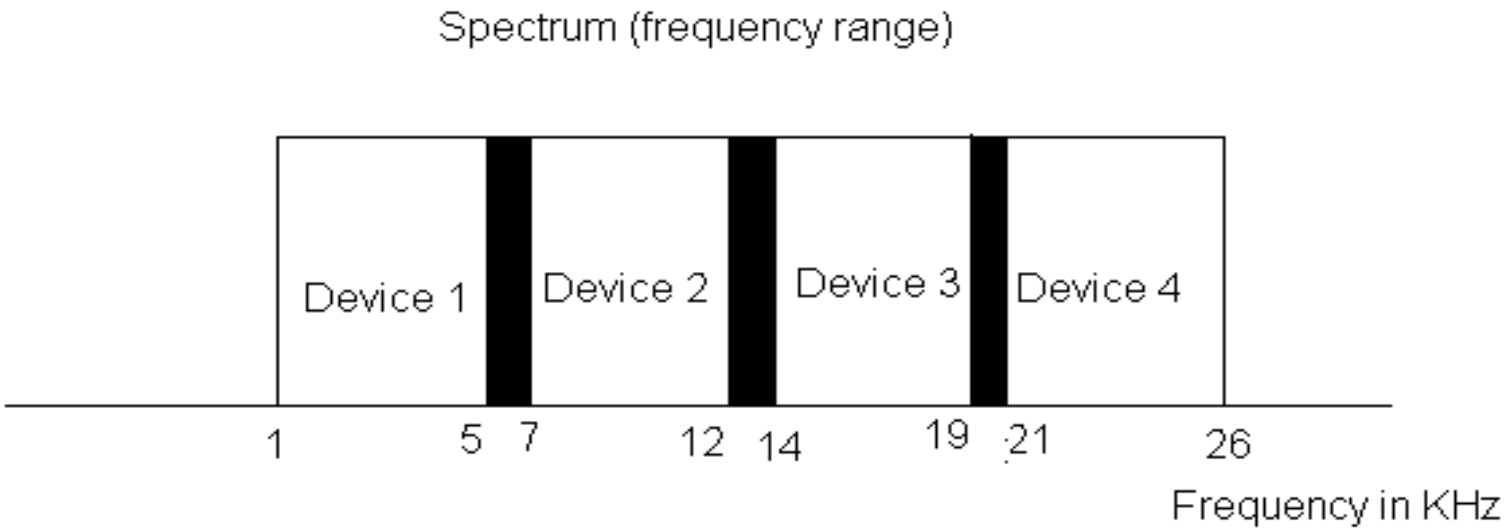


Dividing a link into channels



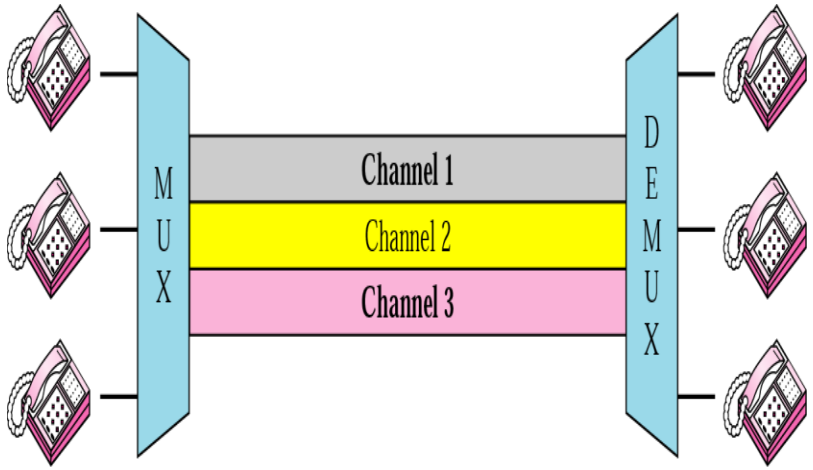
Frequency-division Multiplexing (FDM)

- FDM is an analog technique that can be applied when the bandwidth of a link is greater than the combined bandwidths of the signals to be transmitted.



Frequency-division Multiplexing (FDM)

- In FDM signals generated by each device modulate different carrier frequencies. These modulated signals are combined into a single composite signal that can be transported by the link.



FDM is an analog multiplexing technique that combines signals.

FDM is an analog multiplexing technique that combines analog signals.

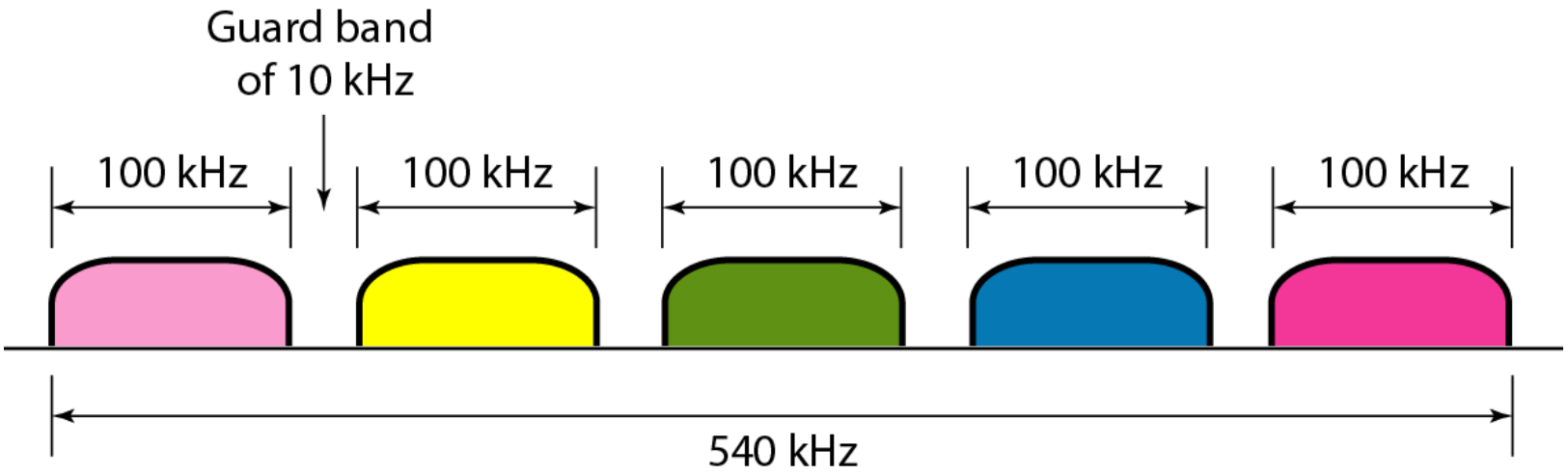
Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in next figure


Guard band
of 10 kHz
100 kHz
100 kHz
100 kHz
100 kHz
100 kHz
540 kHz

Time-division Multiplexing (TDM)

- Time-division multiplexing (TDM) is a digital process that can be applied when the data rate capacity of the transmission medium is greater than the data rate required by the sending and receiving devices.

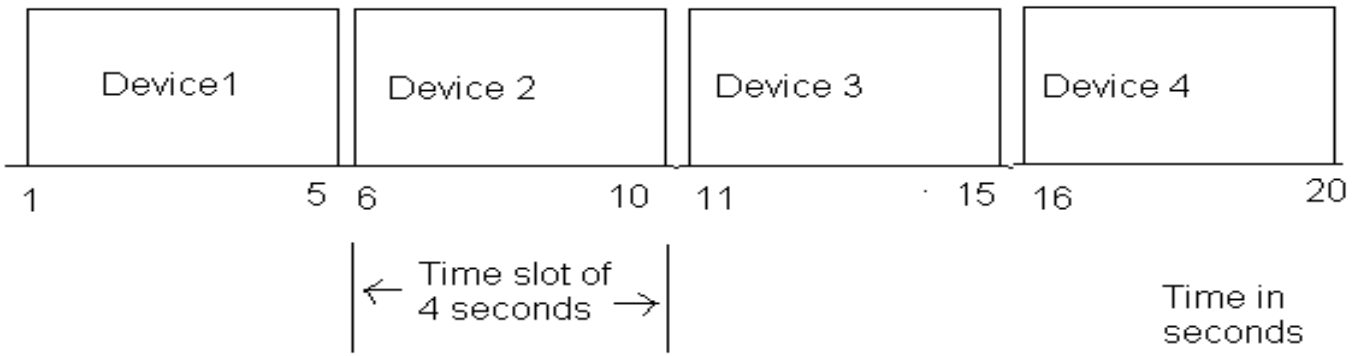
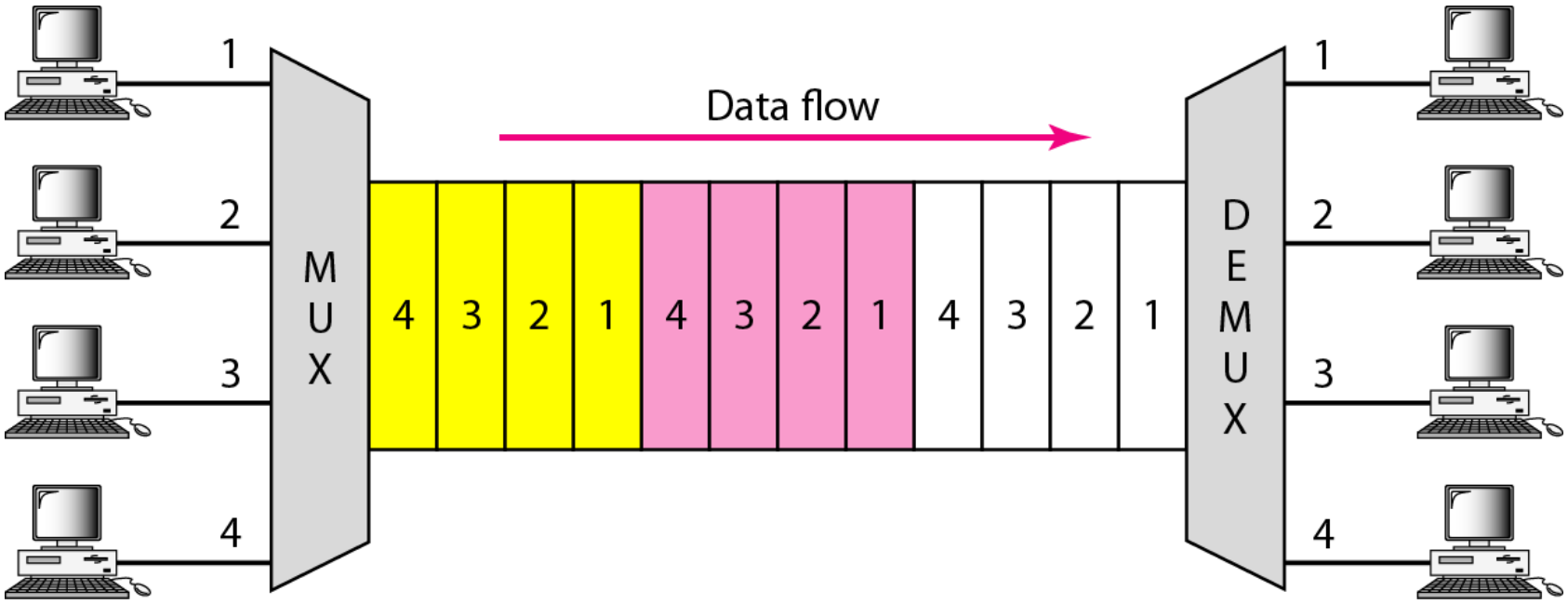


Figure 6.12 *TDM*

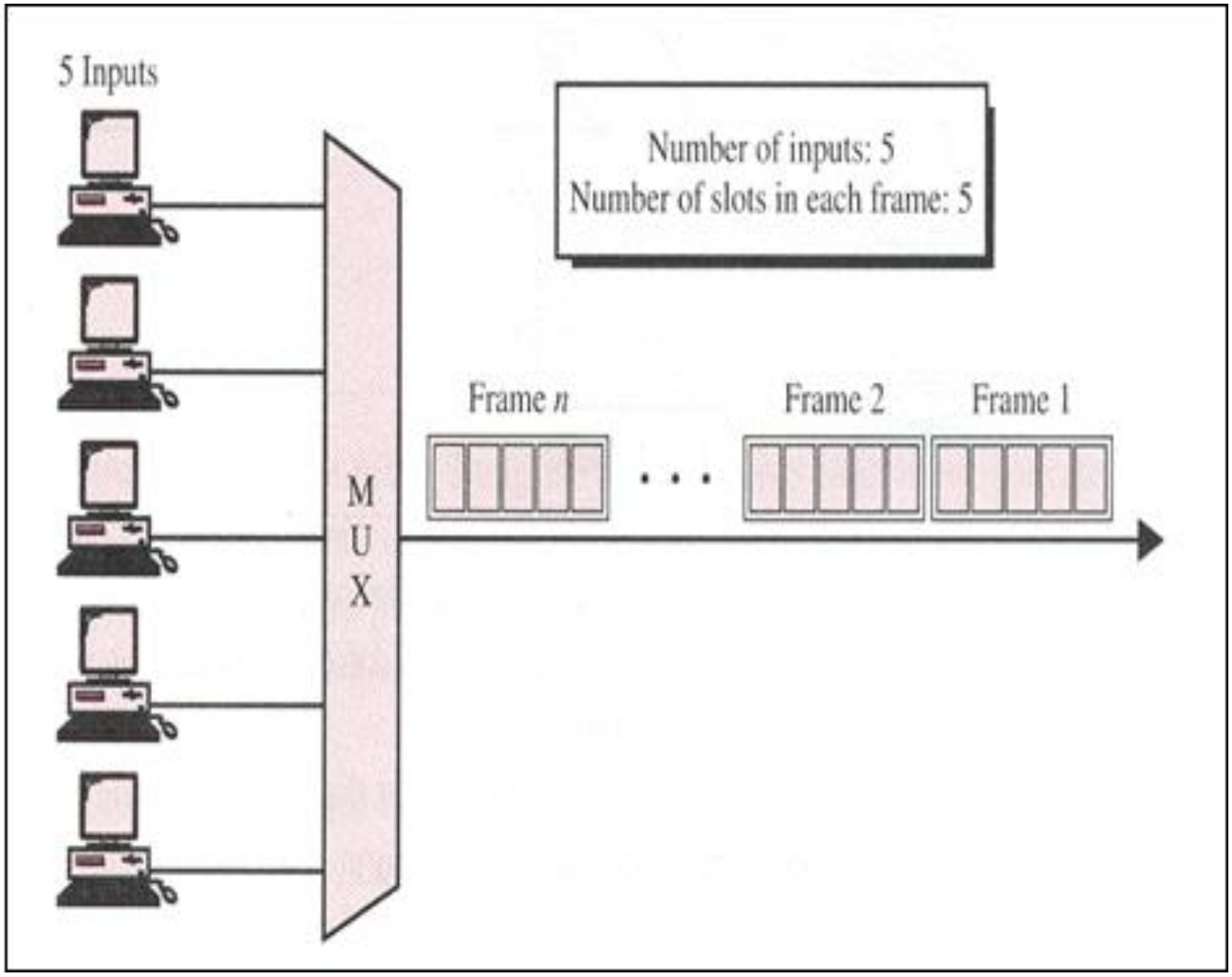


TDM is a digital multiplexing technique for combining several low-rate channels into one high-rate one.

Time-division Multiplexing (TDM)

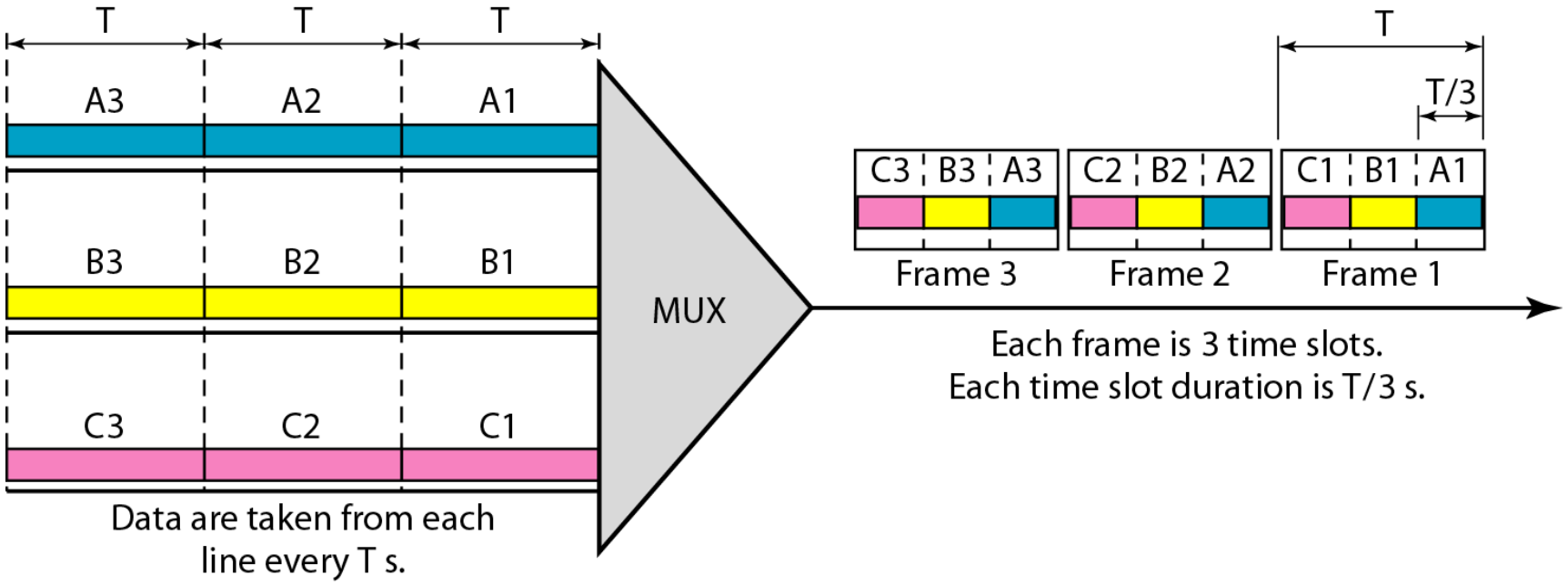
- TDM can be implemented in two ways: synchronous TDM and asynchronous TDM.
- In synchronous time-division multiplexing, the term synchronous means that the multiplexer allocates exactly the same time slot to each device at all times, whether or not a device has anything to transmit.
- Frames

Time slots are grouped into frames. A frame consists of a one complete cycle of time slots, including one or more slots dedicated to each sending device.


5 Inputs
Number of inputs: 5
Number of slots in each frame: 5
Frame n
Frame 2
Frame 1
M
U
X

In synchronous TDM, the data rate of the link is *n* times faster, and the unit duration is *n* times shorter.

Synchronous time-division multiplexing



In Figure shown in previous slide, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

We can answer the questions as follows:

a. The data rate of each input connection is 1 kbps. This means that the bit duration is 1/1000 s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).

b. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is 1/3 ms.

c. Each frame carries three output time slots. So the duration of a frame is 3 × 1/3 ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.

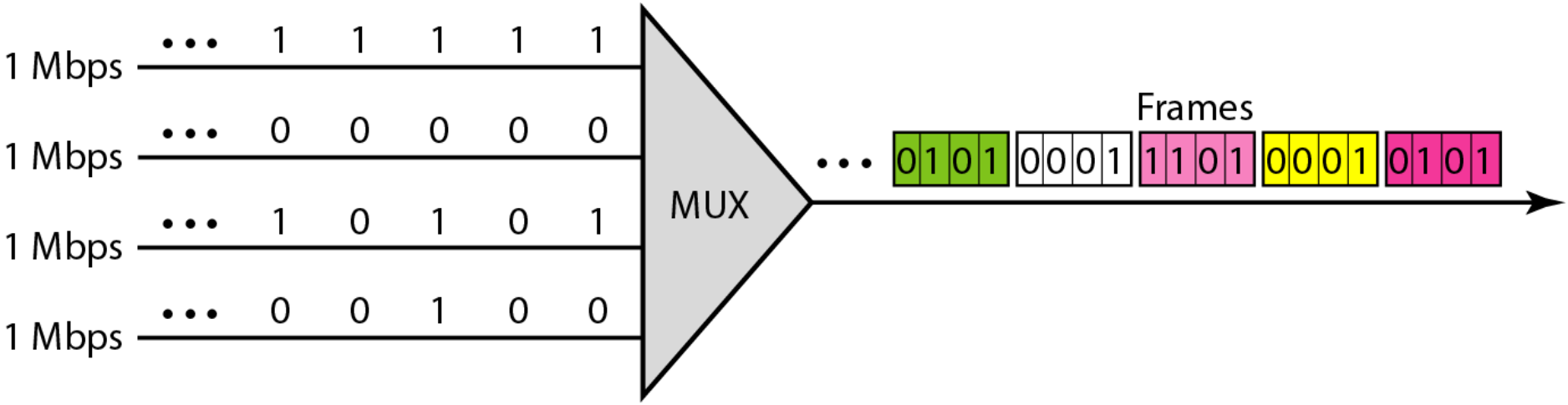

1 Mbps
1 Mbps
1 Mbps
1 Mbps
1 1 1 1 1
0 0 0 0 0
1 0 1 0 1
0 0 1 0 0
MUX
Frames
0101
0001
1101
0001
0101

Figure in previous slide shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

a. The input bit duration is the inverse of the bit rate: 1/1 Mbps = 1 µs.

b. The output bit duration is one-fourth of the input bit duration, or ¼ µs.

c. The output bit rate is the inverse of the output bit duration or 1/(4µs) or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = 4 × 1 Mbps = 4 Mbps.

d. The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame.

Solution

We can answer the questions as follows:

a. The duration of 1 bit before multiplexing is 1 / 1 kbps, or 0.001 s (1 ms).

b. The rate of the link is 4 times the rate of a connection, or 4 kbps.

c. **The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or 1/4 ms or 250 μs. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or 1/4 kbps or 250 μs.**

d. **The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μs, or 1 ms.**